

Multi-Label Classification for Predicting Skin Reactions

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Abstract—This study develops machine learning models to predict skin reactions based on individual characteristics and ingredient concentrations. Random Forest performed best, though class imbalance limited predictions for Positive Reaction. Future work will focus on data expansion and additional predictors to improve model complexity, accuracy, and real-world applicability.

Index Terms—Machine Learning, Skin Care, Random Forest, Multi-Label Classification

I. INTRODUCTION

A. Background and Motivation

The skincare industry is rapidly growing, driven by diverse products and social media influence. However, consumers often struggle to find products suited to their unique skin profiles, leading to adverse reactions such as irritation or ineffective results. To address this, predictive models can provide personalized recommendations, helping users select ingredients that align with their skin characteristics and minimize risks of negative reactions.

B. Study Goal

This research aims to develop machine learning models to predict whether specific skincare ingredients, at certain concentrations, will trigger positive, negative, irritation, brightening, or neutral reactions based on an individual's skin profile, including age, hydration level, melanin concentration, and other measurable attributes.

C. Existing Literature Review

The article, “An Observational Study on Adverse Reactions of Cosmetics: The Need to Practice the Cosmetovigilance System [1],” emphasizes the importance of monitoring adverse reactions to cosmetic products, which is indeed critical to ensuring consumer safety. The researchers focus on cosmetovigilance, a system designed to track such reactions and improve safety protocols. The study found that adverse reactions to cosmetics are prevalent, with skin care, hair care, and makeup products being commonly implicated. The study employed logistic regression as part of its statistical analysis to assess the factors contributing to cosmetics-related adverse events. This method helped identify significant determinants of adverse reactions, such as cosmetic usage patterns, demographics, and prior allergies. This study develops machine learning models to predict skin reactions, building on existing research and advancing personalized skincare and consumer safety.

II. METHOD

A. Problem Formulation

To achieve the goal of predicting skin reactions based on individual characteristics such as age, hydration, and skin type, this study employs several machine learning methods. The dataset used consists of both categorical and continuous variables. The two categorical columns are “Skin Type” and “Ingredient”, with “Skin Type” representing categories like dry and oily, and “Ingredient” including various cosmetic ingredients such as Niacinamide, Vitamin C, and others. The continuous variables include personal skin attributes such as “Melanin Concentration”, “Collagen Amount”, Epidermis Thickness”, “Hydration Level”, and “Concentration” of the ingredient applied.

The following histograms show the distributions of Skin Type, Ingredients, and Reactions in the dataset:

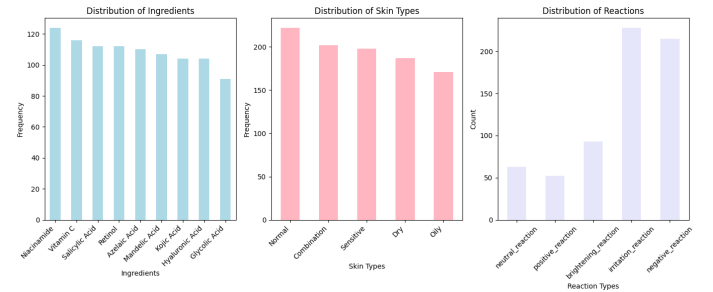


Fig. 1. Data Distribution of Skin Types, Ingredients, and Reactions

In this study, the problem is framed as a multi-label classification task, where the goal is to predict the presence or absence of multiple skin reactions simultaneously. The input features consist of encoded categorical variables (Skin Type and Ingredient) and continuous numerical variables. The models output five binary values, where 1 indicates the presence of a reaction and 0 indicates its absence, corresponding to the five types of skin reactions.

B. Methodologies

To address this problem and identify the best-performing model, various machine learning algorithms were employed, including Logistic Regression, Random Forest, XGBoost, Naive Bayes, and a Neural Network. Each traditional machine

learning model was extended for multi-label prediction using the MultiOutputClassifier, which enables the training of separate binary classifiers for each target label. In contrast, the Neural Network was custom-built with a multi-output architecture, featuring three fully connected layers (128 and 64 neurons) with ReLU and Sigmoid activations. Trained for 100 epochs with Binary Cross-Entropy Loss and a learning rate of 0.001, this algorithm was designed to predict probabilities for all target labels using its multi-output architecture.

III. RESULTS

A. Data Pipeline

The dataset was preprocessed by handling missing values, one-hot encoding categorical features, and converting target reactions to binary. EDA, including distribution analysis and a correlation heatmap, revealed data structures and relationships between features and reactions. After splitting the data into 70% training and 30% testing, models including Logistic Regression, Random Forest, XGBoost, Naive Bayes, and a Neural Network were evaluated using accuracy, Hamming loss, and ROC-AUC scores.

B. Results and Interpretation

A table was generated to summarize the overall performance of different models:

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Hamming Loss
Logistic Regression	0.867347	0.045578
Random Forest	0.928571	0.021769
XGBoost	0.986395	0.004762
Naive Bayes	0.598639	0.227891
Neural Network	0.829932	0.057823

Based on the results in Table I, the XGBoost model achieved the highest performance with an accuracy of 98.64% and the lowest Hamming loss of 0.0048, making it the best-performing model overall. The Random Forest model ranked second with an accuracy of 92.86% and a Hamming loss of 0.0218, indicating strong predictive capability. In contrast, the Naive Bayes model performed the worst, with an accuracy of 59.86% and a significantly higher Hamming loss of 0.2279, highlighting its limitations for this dataset. While the Logistic Regression and Neural Network models showed moderate performance, they were outperformed by XGBoost and Random Forest.

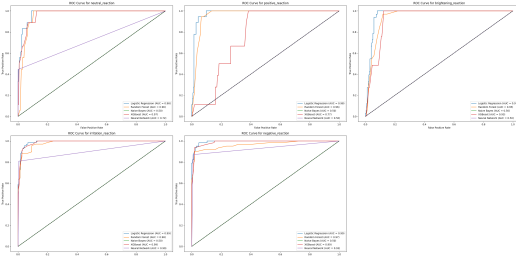


Fig. 2. ROC-AUC curves for all models across different reaction types.

Fig. 2 presents the ROC-AUC curves for all models across the five skin reactions. Logistic Regression and Random Forest demonstrate the strongest performance, achieving consistently high AUC scores (0.95–0.99) across all reactions. However, XGBoost shows noticeably lower performance for Positive Reaction with an AUC of 0.77, indicating its difficulty in distinguishing between positive and negative classes for this label. In contrast, XGBoost performs well for other reactions like Irritation and Negative Reaction, where it achieves near-perfect AUC scores (0.99). The Neural Network delivers moderate performance, excelling for Negative Reaction but underperforming for Positive Reaction and Brightening Reaction. Naive Bayes performs poorly across all reactions, with AUC values around 0.50, reflecting its inability to model meaningful relationships in the data.

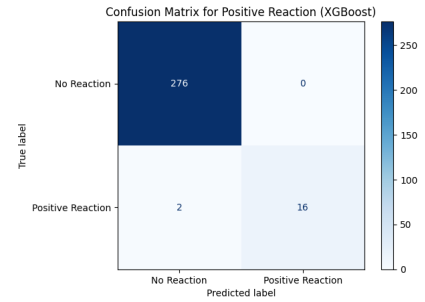


Fig. 3. Confusion Matrix for Positive Reaction (XGBoost).

Based on Fig. 3, the lower AUC scores for Positive Reaction can be attributed to the limited number of positive reaction observations, making this reaction more difficult to predict. Given its consistently strong performance across all reactions, Random Forest is recommended for future implementation as it delivers robust predictions, particularly in scenarios with imbalanced or limited data.

IV. CONCLUSION AND FUTURE WORK

This research developed machine learning models to predict skin reactions using factors such as age, hydration level, skin type, and ingredient concentration. As shown in the table and figures, Random Forest delivered the best performance, achieving high AUC scores, accuracy, and low Hamming loss across all reaction types. However, Positive Reaction was difficult to predict due to data imbalance and a limited number of positive cases.

To improve machine learning predictions in skincare, several enhancements are necessary: increasing data for each reaction type to address class imbalance, incorporating facial images for better analysis with CNN, expanding the ingredient dataset, and accounting for seasonal and environmental factors like temperature and UV exposure. Including age-specific reactions and personal skincare goals will improve accuracy and personalization. These improvements will enhance the robustness, reliability, and real-world applicability of the models in the skincare industry.

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